

SOLIDWORKS API (.NET VBA C#) — Reference

(153/337): IMassProperty2-IMate

SOLIDWORKS API Documentation (converted from CHM)

IMassProperty2 (Type)

IMassProperty2 Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMassProperty2 Interface

Allows access to individual mass properties as found in the Mass Properties dialog box.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IMassProperty2
```

Visual Basic (Usage)

```
Dim instance As IMassProperty2
```

C#

```
public interface IMassProperty2
```

C++/CLI

```
public interface class IMassProperty2
```

Visual Basic for Applications (VBA) Syntax

See [MassProperty2](#).

Example

[Get and Override Mass Properties \(VBA\)](#)
[Get and Override Mass Properties \(VB.NET\)](#)
[Get and Override Mass Properties \(C#\)](#)

Remarks

For more information, see the **Mass Properties Dialog Box** topic in the SOLIDWORKS Design user-interface help.

Accessors

[ModelDocExtension::CreateMassProperty2](#)

Access Diagram

[MassProperty2](#)

See Also

[IMassProperty2 Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IMassProperty2 Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMassProperty2 Interface

For a list of all members of this type, see [IMassProperty2 members](#).

Public Methods

Name	Description
 GetMomentOfInertia	Gets the moment of inertia at the specified coordinate system for the selected bodies/components.
 GetOverrideOptions	Gets the mass property override options for the selected bodies/components.
 Recalculate	Recalculates the mass properties of the selected bodies/components.
 SetCoordinateSystem	Sets the coordinate system to use when calculating mass properties for this model.
 SetOverrideOptions	Sets the mass property override options for the selected bodies/components.

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See Also

[IMassProperty2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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IMassProperty2 Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : IMassProperty2 Interface

The following tables list the members exposed by [IMassProperty2](#).

Public Properties

Name	Description
 AccuracyLevel	Gets or sets the accuracy level used to calculate mass properties.
 CenterOfMass	Gets the center of mass of selected components/bodies.
 Density	Gets the density of selected components/bodies.
 IncludeHiddenBodiesOrComponents	Gets or sets whether to include the mass properties of hidden bodies/components.
 Mass	Gets the mass of selected components/bodies.
 PrincipalAxesOfInertia	Gets the principal axis of inertia for the specified axis.
 PrincipalMomentsOfInertia	Gets the principal moments of inertia.
 SelectedItems	Gets or sets the list of bodies/components for which to calculate mass properties.
 ShowWeldBeadMass	Gets or sets whether to calculate weld bead mass.
 SurfaceArea	Gets the surface area of selected components/bodies.
 UseSystemUnits	Gets or sets whether to use system units or document units when calculating mass properties.
 Volume	Gets the volume of selected components/bodies.
 WeldBeadMass	Gets the weld bead mass.

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Public Methods

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 Recalculate	Recalculates the mass properties of the selected bodies/components.
 SetCoordinateSystem	Sets the coordinate system to use when calculating mass properties for this model.
 SetOverrideOptions	Sets the mass property override options for the selected bodies/components.

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See Also

[IMassProperty2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

IMassProperty2 Interface Properties

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IMassProperty2 Interface

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 PrincipalAxesOfInertia	Gets the principal axis of inertia for the specified axis.
 PrincipalMomentsOfInertia	Gets the principal moments of inertia.
 SelectedItems	Gets or sets the list of bodies/components for which to calculate mass properties.
 ShowWeldBeadMass	Gets or sets whether to calculate weld bead mass.
 SurfaceArea	Gets the surface area of selected components/bodies.
 UseSystemUnits	Gets or sets whether to use system units or document units when calculating mass properties.
 Volume	Gets the volume of selected components/bodies.
 WeldBeadMass	Gets the weld bead mass.

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See Also

[IMassProperty2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

AccuracyLevel Property (IMassProperty2)

[See Also](#)

 

[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : AccuracyLevel Property (IMassProperty2)

Gets or sets the accuracy level used to calculate mass properties.

.NET Syntax

Visual Basic (Declaration)

Property AccuracyLevel As System.Integer

Visual Basic (Usage)



```
Dim instance As IMassProperty2  
Dim value As System.Integer  
  
instance.AccuracyLevel = value  
  
value = instance.AccuracyLevel
```

C#

```
System.int AccuracyLevel {get; set;}
```

C++/CLI

```
property System.int AccuracyLevel {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Accuracy level as defined in [swMassPropertyAccuracyLevel_e](#)

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::AccuracyLevel](#).

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)

Availability

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CenterOfMass Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : CenterOfMass Property (IMassProperty2)

Gets the center of mass of selected components/bodies.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property CenterOfMass As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2
Dim value As System.Object

value = instance.CenterOfMass
```

C#

```
System.Object CenterOfMass {get;}
```

C++/CLI

```
property System.Object^ CenterOfMass {
    System.Object^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of three doubles of the x, y, and z coordinates of the center of mass

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::CenterOfMass](#).

Remarks

If [IMassPropertyOverrideOptions::OverrideCenterOfMass](#) is true or [IMassProperty2::IncludeHiddenBodiesOrComponents](#) is reset, then call [IMassProperty2::Recalculate](#) before using this property.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)
[IMassPropertyOverrideOptions::SetOverrideCenterOfMassValue Method \(\)](#)
[IMassPropertyOverrideOptions::GetOverrideCenterOfMassValue Method \(\)](#)

Availability

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Density Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : Density Property (IMassProperty2)

Gets the density of selected components/bodies.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Density As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2
Dim value As System.Double

value = instance.Density
```

C#

```
System.double Density {get;}
```

C++/CLI

```
property System.double Density {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Density

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::Density](#).

Example

See the [IMassProperty2](#) examples.

Remarks

If the density varies across components in a model:

$$\text{Density} = (\text{Mass} / \text{Volume})$$

If [IMassProperty2::IncludeHiddenBodiesOrComponents](#) is reset, then call [IMassProperty2::Recalculate](#) before using this property.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)

Availability

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GetMomentOfInertia Method (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : GetMomentOfInertia Method (IMassProperty2)

Gets the moment of inertia at the specified coordinate system for the selected bodies/components.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMomentOfInertia( _  
    ByVal WhereTaken As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2  
Dim WhereTaken As System.Integer  
Dim value As System.Object  
  
value = instance.GetMomentOfInertia(WhereTaken)
```

C#

```
System.object GetMomentOfInertia(  
    System.int WhereTaken  
)
```

C++/CLI

```
System.Object^ GetMomentOfInertia(  
& System.int WhereTaken  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhereTaken
Coordinate system as defined in [swMassPropertyMoment_e](#)

Return Value

Array containing the moment of inertia calculations (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::GetMomentOfInertia](#).

Example

See the [IMassProperty2](#) examples.

Remarks

The return value is a 0-based array of nine doubles:

[*Lxx*, *Lxy*, *Lxz*, *Lyx*, *Lyy*, *Lyx*, *Lzx*, *Lzy*, *Lzz*]

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)
[IMassProperty2::PrincipalAxesOfInertia Property \(\)](#)
[IMassProperty2::PrincipalMomentsOfInertia Property \(\)](#)

[IMassPropertyOverrideOptions::SetOverrideMomentsOfInertiaValue Method \(\)](#)
[IMassPropertyOverrideOptions::GetOverrideMomentsOfInertiaValue Method \(\)](#)

Availability

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GetOverrideOptions Method (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : GetOverrideOptions Method (IMassProperty2)

Gets the mass property override options for the selected bodies/components.

.NET Syntax

Visual Basic (Declaration)

```
Function GetOverrideOptions() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2  
Dim value As System.Object  
  
value = instance.GetOverrideOptions()
```

C#

```
System.Object GetOverrideOptions()
```

C++/CLI

```
System.Object^ GetOverrideOptions();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IMassPropertyOverrideOptions](#)

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::GetOverrideOptions](#).

Example

See the [IMassProperty2](#) examples.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)
[IMassProperty2::SetOverrideOptions Method \(\)](#)

Availability

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IncludeHiddenBodiesOrComponents Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : IncludeHiddenBodiesOrComponents Property (IMassProperty2)

Gets or sets whether to include the mass properties of hidden bodies/components.

.NET Syntax

Visual Basic (Declaration)

Property IncludeHiddenBodiesOrComponents As System.Boolean

Visual Basic (Usage)



```
Dim instance As IMassProperty2
Dim value As System.Boolean

instance.IncludeHiddenBodiesOrComponents = value

value = instance.IncludeHiddenBodiesOrComponents
```

C#

```
System.bool IncludeHiddenBodiesOrComponents {get; set;}
```

C++/CLI

```
property System.bool IncludeHiddenBodiesOrComponents {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to include the mass properties of hidden bodies/components, false to not

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::IncludeHiddenBodiesOrComponents](#).

Remarks

After setting this property to true, be sure to call [IMassProperty2::Recalculate](#) before using the IMassProperty2 properties.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)
[IMassProperty2::SelectedItems Property\(\)](#)

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Mass Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : Mass Property (IMassProperty2)

Gets the mass of selected components/bodies.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Mass As System.Double

Visual Basic (Usage)



```
Dim instance As IMassProperty2
Dim value As System.Double

value = instance.Mass
```

C#

```
System.double Mass {get;}
```

C++/CLI

```
property System.double Mass {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Mass

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::Mass](#).

Example

See the [IMassProperty2](#) examples.

Remarks

If [IMassPropertyOverrideOptions::OverrideMass](#) is true or [IMassProperty2::IncludeHiddenBodiesOrComponents](#) is reset, then call [IMassProperty2::Recalculate](#) before using this property.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)
[IMassPropertyOverrideOptions::SetOverrideMassValue Method \(\)](#)
[IMassPropertyOverrideOptions::GetOverrideMassValue Method \(\)](#)

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PrincipalAxesOfInertia Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : PrincipalAxesOfInertia Property (IMassProperty2)

Gets the principal axis of inertia for the specified axis.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property PrincipalAxesOfInertia( _  
    ByVal Axis As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2  
Dim Axis As System.Integer  
Dim value As System.Object
```

```
value = instance.PrincipalAxesOfInertia(Axis)
```

C#

```
System.object PrincipalAxesOfInertia(  
    System.int Axis  
) {get;}
```

C++/CLI

```
property System.Object^ PrincipalAxesOfInertia {  
    System.Object^ get(System.int Axis);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Axis

- 0 = x axis
- 1 = y axis
- 2 = z axis

Property Value

Zero-based array of size 3 (see [Remarks](#))

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::PrincipalAxesOfInertia](#).

Remarks

This method returns:

- A vector that represents the specified Axis: lx, ly, or lz.
- Metric units unless explicitly specified otherwise.

If [IMassPropertyOverrideOptions::OverrideMomentsOfInertia](#) is true or [IMassProperty2::IncludeHiddenBodiesOrComponents](#) is reset, then call [IMassProperty2::Recalculate](#) before using this property.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)
[IMassPropertyOverrideOptions::SetOverridePrincipalAxesOrientation Method \(\)](#)
[IMassProperty2::PrincipalMomentsOfInertia Property \(\)](#)
[IMassProperty2::GetMomentOfInertia Method \(\)](#)
[IMassPropertyOverrideOptions::GetOverridePrincipalAxesOrientation Method \(\)](#)

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PrincipalMomentsOfInertia Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : PrincipalMomentsOfInertia Property (IMassProperty2)

Gets the principal moments of inertia.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property PrincipalMomentsOfInertia As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2  
Dim value As System.Object
```

```
value = instance.PrincipalMomentsOfInertia
```

C#

```
System.object PrincipalMomentsOfInertia {get;}
```

C++/CLI

```
property System.Object^ PrincipalMomentsOfInertia {  
    System.Object^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of size 3 (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::PrincipalMomentsOfInertia](#).

Example

See the [IMassProperty2](#) examples.

Remarks

This property returns metric units unless explicitly specified otherwise.

The return value is a 0-based array of doubles:

[*P_x*, *P_y*, *P_z*]

If [IMassPropertyOverrideOptions::OverrideMomentsOfInertia](#) is true or [IMassProperty2::IncludeHiddenBodiesOrComponents](#) is reset, then call [IMassProperty2::Recalculate](#) before using this property.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)
[IMassPropertyOverrideOptions::SetOverrideMomentsOfInertiaValue Method \(\)](#)
[IMassPropertyOverrideOptions::SetOverridePrincipalMomentsOfInertia Method \(\)](#)
[IMassProperty2::GetMomentOfInertia Method \(\)](#)
[IMassProperty2::PrincipalAxesOfInertia Property \(\)](#)
[IMassPropertyOverrideOptions::GetOverridePrincipalMomentsOfInertia Method \(\)](#)
[IMassPropertyOverrideOptions::GetOverrideMomentsOfInertiaValue Method \(\)](#)

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Recalculate Method (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : Recalculate Method (IMassProperty2)

Recalculates the mass properties of the selected bodies/components.

.NET Syntax

Visual Basic (Declaration)

```
Function Recalculate() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2  
Dim value As System.Boolean
```

```
value = instance.Recalculate()
```

C#

```
System.bool Recalculate()
```

C++/CLI

```
System.bool Recalculate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if recalculation is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::Recalculate](#).

Example

See the [IMassProperty2](#) examples.

Remarks

Call this method after overriding mass properties (center of mass, mass, moments of inertia, principal axes orientation, or principal moments of inertia) or resetting [IMassProperty2::IncludeHiddenBodiesOrComponents](#).

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)

Availability

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SelectedItems Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : SelectedItems Property (IMassProperty2)

Gets or sets the list of bodies/components for which to calculate mass properties.

.NET Syntax

Visual Basic (Declaration)

```
Property SelectedItems As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2  
Dim value As System.Object
```

```
instance.SelectedItems = value
```

```
value = instance.SelectedItems
```

C#

```
System.Object SelectedItems {get; set;}
```

C++/CLI

```
property System.Object^ SelectedItems {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [bodies](#) for the part or [components](#) for the assembly

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::SelectedItems](#).

Remarks

Pre-selected bodies/components are included in the returned array.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)
[IMassProperty2::IncludeHiddenBodiesOrComponents Property\(\)](#)

Availability

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SetCoordinateSystem Method (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : SetCoordinateSystem Method (IMassProperty2)

Sets the coordinate system to use when calculating mass properties for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function SetCoordinateSystem(  
    ByVal Coords As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2  
Dim Coords As System.Object  
Dim value As System.Boolean  
  
value = instance.SetCoordinateSystem(Coords)
```

C#

```
System.bool SetCoordinateSystem(  
    System.object Coords  
)
```

C++/CLI

```
System.bool SetCoordinateSystem(  
& System.Object^ Coords  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Coords
[Math transform](#)

Return Value

True if the coordinate system is set, false if not

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::SetCoordinateSystem](#).

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)

Availability

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SetOverrideOptions Method (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : SetOverrideOptions Method (IMassProperty2)

Sets the mass property override options for the selected bodies/components.

.NET Syntax

Visual Basic (Declaration)

```
Function SetOverrideOptions( _  
    ByVal Options As System.Object, _  
    ByVal Config\_option As System.Integer, _  
    ByVal Config\_names As System.Object _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2  
Dim Options As System.Object  
Dim Config_option As System.Integer  
Dim Config_names As System.Object  
Dim value As System.Boolean  
  
value = instance.SetOverrideOptions(Options, Config_option, Config_names)
```

C#

```
System.bool SetOverrideOptions(  
    System.object Options,  
    System.int Config\_option,  
    System.object Config\_names  
)
```

C++/CLI

```
System.bool SetOverrideOptions(  
& System.Object^ Options,  
& System.int Config\_option,  
& System.Object^ Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Options

[IMassPropertyOverrideOptions](#)

Config_option

Configuration option as defined in [swInConfigurationOptions_e](#)

Config_names

Array of configuration names; valid only if Config_option = swInConfigurationOptions_e.swSpecifyConfiguration

Return Value

True if the mass property override options are set successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::SetOverrideOptions](#).

Example

See the [IMassProperty2](#) examples.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)
[IMassProperty2::GetOverrideOptions Method \(\)](#)

Availability

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ShowWeldBeadMass Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : ShowWeldBeadMass Property (IMassProperty2)

Gets or sets whether to calculate weld bead mass.

.NET Syntax

Visual Basic (Declaration)

```
Property ShowWeldBeadMass As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2  
Dim value As System.Boolean  
  
instance.ShowWeldBeadMass = value  
  
value = instance.ShowWeldBeadMass
```

C#

```
System.bool ShowWeldBeadMass {get; set;}
```

C++/CLI

```
property System.bool ShowWeldBeadMass {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to calculate weld bead mass, false to not

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::ShowWeldBeadMass](#).

Remarks

After setting this property to true, you can use [IMassProperty2::WeldBeadMass](#).

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)

Availability

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SurfaceArea Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : SurfaceArea Property (IMassProperty2)

Gets the surface area of selected components/bodies.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property SurfaceArea As System.Double

Visual Basic (Usage)



```
Dim instance As IMassProperty2
Dim value As System.Double
```

```
value = instance.SurfaceArea
```

C#

```
System.double SurfaceArea {get;}
```

C++/CLI

```
property System.double SurfaceArea {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Surface area

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::SurfaceArea](#).

Example

See the [IMassProperty2](#) examples.

Remarks

If [IMassProperty2::IncludeHiddenBodiesOrComponents](#) is reset, then call [IMassProperty2::Recalculate](#) before using this property.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)

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UseSystemUnits Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : UseSystemUnits Property (IMassProperty2)

Gets or sets whether to use system units or document units when calculating mass properties.

.NET Syntax

Visual Basic (Declaration)

```
Property UseSystemUnits As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassProperty2
Dim value As System.Boolean

instance.UseSystemUnits = value
value = instance.UseSystemUnits
```

C#

```
System.bool UseSystemUnits {get; set;}
```

C++/CLI

```
property System.bool UseSystemUnits {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use system units, false to use document units

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::UseSystemUnits](#).

Example

See the [IMassProperty2](#) examples.

Remarks

The default value is true. Thus, system units (meters, radians, and kilograms) are used by default. All properties returning a value are adjusted accordingly.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)

Availability

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SOLIDWORKS API Help

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Volume Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : Volume Property (IMassProperty2)

Gets the volume of selected components/bodies.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Volume As System.Double
```

Visual Basic (Usage)

```
Dim instance As IMassProperty2
Dim value As System.Double

value = instance.Volume
```

C#

```
System.double Volume {get;}
```

C++/CLI

```
property System.double Volume {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Volume

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::Volume](#).

Example

See the [IMassProperty2](#) examples.

Remarks

If [IMassProperty2::IncludeHiddenBodiesOrComponents](#) is reset, then call [IMassProperty2::Recalculate](#) before using this property.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)

Availability

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SOLIDWORKS API Help

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WeldBeadMass Property (IMassProperty2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassProperty2 Interface](#) : WeldBeadMass Property (IMassProperty2)

Gets the weld bead mass.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property WeldBeadMass As System.Double
```

Visual Basic (Usage)

```
Dim instance As IMassProperty2
Dim value As System.Double

value = instance.WeldBeadMass
```

C#

```
System.double WeldBeadMass {get;}
```

```
property System.double WeldBeadMass {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Weld bead mass

Visual Basic for Applications (VBA) Syntax

See [MassProperty2::WeldBeadMass](#).

Remarks

This property is valid only if [IMassProperty2::ShowWeldBeadMass](#) is set to true.

See Also

[IMassProperty2 Interface](#)
[IMassProperty2 Members](#)

Availability

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IMassPropertyOverrideOptions (Type)

IMassPropertyOverrideOptions Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMassPropertyOverrideOptions Interface

Allows access to mass property override options.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IMassPropertyOverrideOptions
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions
```

C#

```
public interface IMassPropertyOverrideOptions
```

C++/CLI

```
public interface class IMassPropertyOverrideOptions
```

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions](#).

Example

See the [IMassProperty2](#) examples.

Remarks

Mass property override options are available for only one selected body/component at a time.
For more information, see the **Mass Properties Dialog Box** topic in the SOLIDWORKS Design user-interface help.

Accessors

[IMassProperty2::GetOverrideOptions](#)

Access Diagram

[MassPropertyOverrideOptions](#)

See Also

[IMassPropertyOverrideOptions Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IMassPropertyOverrideOptions Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMassPropertyOverrideOptions Interface

For a list of all members of this type, see [IMassPropertyOverrideOptions members](#).

Public Methods

Name	Description
 GetOverrideCenterOfMassValue	Gets the override center of mass.
 GetOverrideMassValue	Gets the override mass.
 GetOverrideMomentsOfInertiaValue	Gets the override moments of inertia.
 GetOverridePrincipalAxesOrientation	Gets the override principal axis of inertia.
 GetOverridePrincipalMomentsOfInertia	Gets the override principal moments of inertia.
 SetOverrideCenterOfMassValue	Overrides the calculated center of mass of the model currently being edited.
 SetOverrideMassValue	Overrides the calculated mass of the model currently being edited.
 SetOverrideMomentsOfInertiaValue	Overrides the calculated moments of inertia of the model currently being edited.
 SetOverridePrincipalAxesOrientation	Overrides the orientation of the specified principal axis of inertia of the model currently being edited.
 SetOverridePrincipalMomentsOfInertia	Overrides the calculated principal moments of inertia of the model currently being edited.

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See Also

[IMassPropertyOverrideOptions Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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IMassPropertyOverrideOptions Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : IMassPropertyOverrideOptions Interface

The following tables list the members exposed by [IMassPropertyOverrideOptions](#).

Public Properties

Name	Description
 OverrideCenterOfMass	Gets or sets whether to override the calculated center of mass for the model currently being edited.
 OverrideMass	Gets or sets whether to override the calculated mass value for the model currently being edited.
 OverrideMomentsOfInertia	Gets or sets whether to override the calculated moments of inertia for the model currently being edited.

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Public Methods

Name	Description
 GetOverrideCenterOfMassValue	Gets the override center of mass.
 GetOverrideMassValue	Gets the override mass.
 GetOverrideMomentsOfInertiaValue	Gets the override moments of inertia.
 GetOverridePrincipalAxesOrientation	Gets the override principal axis of inertia.
 GetOverridePrincipalMomentsOfInertia	Gets the override principal moments of inertia.
 SetOverrideCenterOfMassValue	Overrides the calculated center of mass of the model currently being edited.
 SetOverrideMassValue	Overrides the calculated mass of the model currently being edited.
 SetOverrideMomentsOfInertiaValue	Overrides the calculated moments of inertia of the model currently being edited.
 SetOverridePrincipalAxesOrientation	Overrides the orientation of the specified principal axis of inertia of the model currently being edited.
 SetOverridePrincipalMomentsOfInertia	Overrides the calculated principal moments of inertia of the model currently being edited.

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See Also

[IMassPropertyOverrideOptions Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

IMassPropertyOverrideOptions Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IMassPropertyOverrideOptions Interface

For a list of all members of this type, see [IMassPropertyOverrideOptions members](#).

Public Properties

Name	Description
 OverrideCenterOfMass	Gets or sets whether to override the calculated center of mass for the model currently being edited.
 OverrideMass	Gets or sets whether to override the calculated mass value for the model currently being edited.
 OverrideMomentsOfInertia	Gets or sets whether to override the calculated moments of inertia for the model currently being edited.

[Top](#)

See Also

[IMassPropertyOverrideOptions Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

GetOverrideCenterOfMassValue Method (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : GetOverrideCenterOfMassValue Method (IMassPropertyOverrideOptions)

Gets the override center of mass.

.NET Syntax

Visual Basic (Declaration)

```
Function GetOverrideCenterOfMassValue( _  
    ByRef CoordinateSystemName As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions  
Dim CoordinateSystemName As System.String  
Dim value As System.Object  
  
value = instance.GetOverrideCenterOfMassValue(CoordinateSystemName)
```

C#

```
System.object GetOverrideCenterOfMassValue(  
    out System.string CoordinateSystemName  
)
```

C++/CLI

```
System.Object^ GetOverrideCenterOfMassValue(  
& [Out] System.String^ CoordinateSystemName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CoordinateSystemName
Name of the coordinate system in which the center of mass is defined

Return Value

Array of three doubles of the x, y, and z coordinates of the center of mass

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::GetOverrideCenterOfMassValue](#).

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassPropertyOverrideOptions::SetOverrideCenterOfMassValue Method \(\)](#)
[IMassProperty2::CenterOfMass Property \(\)](#)
[IMassPropertyOverrideOptions::OverrideCenterOfMass Property \(\)](#)

Availability

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SOLIDWORKS API Help

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GetOverrideMassValue Method (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : GetOverrideMassValue Method (IMassPropertyOverrideOptions)

Gets the override mass.

.NET Syntax

Visual Basic (Declaration)

```
Function GetOverrideMassValue() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions  
Dim value As System.Double
```

```
value = instance.GetOverrideMassValue()
```

C#

```
System.double GetOverrideMassValue()
```

C++/CLI

```
System.double GetOverrideMassValue();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Override mass value

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::GetOverrideMassValue](#).

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassPropertyOverrideOptions::SetOverrideMassValue Method \(\)](#)
[IMassPropertyOverrideOptions::OverrideMass Property \(\)](#)
[IMassProperty2::Mass Property \(\)](#)

Availability

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GetOverrideMomentsOfInertiaValue Method (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : GetOverrideMomentsOfInertiaValue Method (IMassPropertyOverrideOptions)

Gets the override moments of inertia.

.NET Syntax

Visual Basic (Declaration)

```
Function GetOverrideMomentsOfInertiaValue( _  
    ByRef IsReferenceFrameSet As System.Boolean, _  
    ByRef CoordinateSystemName As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions  
Dim IsReferenceFrameSet As System.Boolean  
Dim CoordinateSystemName As System.String  
Dim value As System.Object  
  
value = instance.GetOverrideMomentsOfInertiaValue(IsReferenceFrameSet, CoordinateSystemName)
```

C#

```
System.Object GetOverrideMomentsOfInertiaValue(  
    out System.bool IsReferenceFrameSet,  
    out System.string CoordinateSystemName  
)
```

C++/CLI

```
System.Object^ GetOverrideMomentsOfInertiaValue(  
& [Out] System.bool IsReferenceFrameSet,  
& [Out] System.String^ CoordinateSystemName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IsReferenceFrameSet
True if reference frame is set, false if not
CoordinateSystemName
Name of coordinate system

Return Value

Array of nine doubles: [Lxx, Lxy, Lxz, Lyx, Lyy, Lyz, Lzx, Lzy, Lzz]

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::GetOverrideMomentsOfInertiaValue](#).

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassPropertyOverrideOptions::SetOverrideMomentsOfInertiaValue Method \(\)](#)
[IMassPropertyOverrideOptions::OverrideMomentsOfInertia Property \(\)](#)
[IMassProperty2::GetMomentOfInertia Method \(\)](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

GetOverridePrincipalAxesOrientation Method (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : GetOverridePrincipalAxesOrientation Method (IMassPropertyOverrideOptions)

Gets the override principal axis of inertia.

.NET Syntax

Visual Basic (Declaration)

```
Function GetOverridePrincipalAxesOrientation( _  
    ByVal Axis As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions  
Dim Axis As System.Integer  
Dim value As System.Object  
  
value = instance.GetOverridePrincipalAxesOrientation(Axis)
```

C#

```
System.Object GetOverridePrincipalAxesOrientation(  
    System.int Axis  
)
```

C++/CLI

```
System.Object^ GetOverridePrincipalAxesOrientation(  
    & System.int Axis  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Axis

One of the following principal axes:

- 0 = X axis
- 1 = Y axis
- 2 = Z axis

Return Value

An array of three doubles of the x, y, and z coordinates of Axis

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::GetOverridePrincipalAxesOrientation](#).

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassPropertyOverrideOptions::SetOverridePrincipalAxesOrientation Method \(\)](#)
[IMassPropertyOverrideOptions::OverrideMomentsOfInertia Property \(\)](#)
[IMassProperty2::PrincipalAxesOfInertia Property \(\)](#)

Availability

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SOLIDWORKS API Help

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GetOverridePrincipalMomentsOfInertia Method (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : GetOverridePrincipalMomentsOfInertia Method (IMassPropertyOverrideOptions)

Gets the override principal moments of inertia.

.NET Syntax

Visual Basic (Declaration)

```
Function GetOverridePrincipalMomentsOfInertia() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions
Dim value As System.Object

value = instance.GetOverridePrincipalMomentsOfInertia()
```

C#

```
System.Object GetOverridePrincipalMomentsOfInertia()
```

C++/CLI

```
System.Object^ GetOverridePrincipalMomentsOfInertia();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of three doubles of the principal moments of inertia: [Px, Py, Pz]

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::GetOverridePrincipalMomentsOfInertia](#).

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassPropertyOverrideOptions::SetOverridePrincipalMomentsOfInertia Method \(\)](#)
[IMassPropertyOverrideOptions::OverrideMomentsOfInertia Property \(\)](#)
[IMassProperty2::PrincipalMomentsOfInertia Property \(\)](#)

Availability

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SOLIDWORKS API Help

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OverrideCenterOfMass Property (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : OverrideCenterOfMass Property (IMassPropertyOverrideOptions)

Gets or sets whether to override the calculated center of mass for the model currently being edited.

.NET Syntax

Visual Basic (Declaration)

```
Property OverrideCenterOfMass As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions
Dim value As System.Boolean
```

```
instance.OverrideCenterOfMass = value
```

```
value = instance.OverrideCenterOfMass
```

C#

```
System.bool OverrideCenterOfMass {get; set;}
```

C++/CLI

```
property System.bool OverrideCenterOfMass {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to override the calculated center of mass, false to not

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::OverrideCenterOfMass](#).

Remarks

After setting this property to true, call [IMassPropertyOverrideOptions::SetOverrideCenterOfMassValue](#).

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassProperty2::CenterOfMass Property \(\)](#)
[IMassPropertyOverrideOptions::GetOverrideCenterOfMassValue Method \(\)](#)

Availability

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SOLIDWORKS API Help

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OverrideMass Property (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : OverrideMass Property (IMassPropertyOverrideOptions)

Gets or sets whether to override the calculated mass value for the model currently being edited.

.NET Syntax

Visual Basic (Declaration)

```
Property OverrideMass As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions  
Dim value As System.Boolean
```

```
instance.OverrideMass = value
```

```
value = instance.OverrideMass
```

C#

```
System.bool OverrideMass {get; set;}
```

C++/CLI

```
property System.bool OverrideMass {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to override the calculated mass value, false to not

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::OverrideMass](#).

Example

See the [IMassProperty2](#) examples.

Remarks

After setting this property to true, call [IMassPropertyOverrideOptions::SetOverrideMassValue](#).

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassProperty2::Mass Property \(\)](#)
[IMassPropertyOverrideOptions::GetOverrideMassValue Method \(\)](#)

Availability

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SOLIDWORKS API Help

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OverrideMomentsOfInertia Property (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : OverrideMomentsOfInertia Property (IMassPropertyOverrideOptions)

Gets or sets whether to override the calculated moments of inertia for the model currently being edited.

.NET Syntax

Visual Basic (Declaration)

```
Property OverrideMomentsOfInertia As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions  
Dim value As System.Boolean
```

```
instance.OverrideMomentsOfInertia = value
```

```
value = instance.OverrideMomentsOfInertia
```

C#

```
System.bool OverrideMomentsOfInertia {get; set;}
```

C++/CLI

```
property System.bool OverrideMomentsOfInertia {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to override the calculated moments of inertia, false to not

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::OverrideMomentsOfInertia](#).

Example

See the [IMassProperty2](#) examples.

Remarks

After setting this property to true, call:

- [IMassPropertyOverrideOptions::SetOverrideMomentsOfInertiaValue](#)
- [IMassPropertyOverrideOptions::SetOverridePrincipalAxesOrientation](#)
- [IMassPropertyOverrideOptions::SetOverridePrincipalMomentsOfInertia](#)

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassProperty2::GetMomentOfInertia Method \(\)](#)
[IMassProperty2::PrincipalAxesOfInertia Property \(\)](#)
[IMassProperty2::PrincipalMomentsOfInertia Property \(\)](#)
[IMassPropertyOverrideOptions::GetOverrideMomentsOfInertiaValue Method \(\)](#)
[IMassPropertyOverrideOptions::GetOverridePrincipalMomentsOfInertia Method \(\)](#)
[IMassPropertyOverrideOptions::GetOverridePrincipalAxesOrientation Method \(\)](#)

Availability

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[SOLIDWORKS API Help](#)

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SetOverrideCenterOfMassValue Method (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : SetOverrideCenterOfMassValue Method (IMassPropertyOverrideOptions)

Overrides the calculated center of mass of the model currently being edited.

.NET Syntax

Visual Basic (Declaration)

```
Function SetOverrideCenterOfMassValue( _  
    ByVal Value As System.Object, _  
    ByVal CoordinateSystemName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions  
Dim Value As System.Object  
Dim CoordinateSystemName As System.String  
Dim value As System.Boolean  
  
value = instance.SetOverrideCenterOfMassValue(Value, CoordinateSystemName)
```

C#

```
System.bool SetOverrideCenterOfMassValue(  
    System.object Value,  
    System.string CoordinateSystemName  
)
```

C++/CLI

```
System.bool SetOverrideCenterOfMassValue(  
& System.Object^ Value,  
& System.String^ CoordinateSystemName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Value

Array of three doubles of the x, y, and z coordinates of the center of mass

CoordinateSystemName

Name of the coordinate system in which the center of mass is defined

Return Value

True if the center of mass is successfully overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::SetOverrideCenterOfMassValue](#).

Remarks

This method is valid only if [IMassPropertyOverrideOptions::OverrideCenterOfMass](#) is set to true.

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassProperty2::CenterOfMass Property \(\)](#)

Availability

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SOLIDWORKS API Help

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SetOverrideMassValue Method (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : SetOverrideMassValue Method (IMassPropertyOverrideOptions)

Overrides the calculated mass of the model currently being edited.

.NET Syntax

Visual Basic (Declaration)

```
Function SetOverrideMassValue( _  
    ByVal Value As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions  
Dim Value As System.Double  
Dim value As System.Boolean  
  
value = instance.SetOverrideMassValue(Value)
```

C#

```
System.bool SetOverrideMassValue(  
    System.double Value  
)
```

C++/CLI

```
System.bool SetOverrideMassValue(  
& System.double Value  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Value
New mass value

Return Value

True if the mass is successfully overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::SetOverrideMassValue](#).

Example

See the [IMassProperty2](#) examples.

Remarks

This method is valid only if [IMassPropertyOverrideOptions::OverrideMass](#) is set to true.

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassProperty2::Mass Property \(\)](#)

Availability

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SOLIDWORKS API Help

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SetOverrideMomentsOfInertiaValue Method (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : SetOverrideMomentsOfInertiaValue Method (IMassPropertyOverrideOptions)

Overrides the calculated moments of inertia of the model currently being edited.

.NET Syntax

Visual Basic (Declaration)

```
Function SetOverrideMomentsOfInertiaValue( _  
    ByVal ReferenceFrame As System.Integer, _  
    ByVal Value As System.Object, _  
    ByVal CoordinateSystemName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions  
Dim ReferenceFrame As System.Integer  
Dim Value As System.Object  
Dim CoordinateSystemName As System.String  
Dim value As System.Boolean
```

```
value = instance.SetOverrideMomentsOfInertiaValue(ReferenceFrame, Value, CoordinateSystemName)
```

C#

```
System.bool SetOverrideMomentsOfInertiaValue(  
    System.int ReferenceFrame,  
    System.object Value,  
    System.string CoordinateSystemName  
)
```

C++/CLI

```
System.bool SetOverrideMomentsOfInertiaValue(  
& System.int ReferenceFrame,  
& System.Object^ Value,  
& System.String^ CoordinateSystemName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ReferenceFrame

Frame of reference as defined in [swMomentsOfInertiaReferenceFrame_e](#)

Value

Array of nine doubles: [*Lxx*, *Lxy*, *Lxz*, *Lyx*, *Lyx*, *Lyz*, *Lzx*, *Lzy*, *Lzz*]

CoordinateSystemName

Name of coordinate system; valid only if ReferenceFrame = swMomentsOfInertiaReferenceFrame_e.swMomentsOfInertiaReferenceFrame_UserCoordinateSystem

Return Value

True if the moments of inertia are successfully overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::SetOverrideMomentsOfInertiaValue](#).

Example

See the [IMassProperty2](#) examples.

Remarks

This method is valid only if [IMassPropertyOverrideOptions::OverrideMomentsOfInertia](#) is set to true.

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassProperty2::GetMomentOfInertia Method \(\)](#)

Availability

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SOLIDWORKS API Help

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SetOverridePrincipalAxesOrientation Method (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : SetOverridePrincipalAxesOrientation Method (IMassPropertyOverrideOptions)

Overrides the orientation of the specified principal axis of inertia of the model currently being edited.

.NET Syntax

Visual Basic (Declaration)

```
Function SetOverridePrincipalAxesOrientation( _  
    ByVal Axis As System.Integer, _  
    ByVal Value As System.Object, _  
    ByVal AutoCorrect As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions  
Dim Axis As System.Integer  
Dim Value As System.Object  
Dim AutoCorrect As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SetOverridePrincipalAxesOrientation(Axis, Value, AutoCorrect)
```

C#

```
System.bool SetOverridePrincipalAxesOrientation(  
    System.int Axis,  
    System.object Value,  
    System.bool AutoCorrect  
)
```

C++/CLI

```
System.bool SetOverridePrincipalAxesOrientation(  
& System.int Axis,  
& System.Object^ Value,  
& System.bool AutoCorrect  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Axis

One of the following principal axes:

- 0 = X axis

- 1 = Y axis
- 2 = Z axis

Value

An array of three doubles of the x, y, and z coordinates of Axis

AutoCorrect

True to generate orthogonal axes, false to not

Return Value

True if the principal axis orientation successfully overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::SetOverridePrincipalAxesOrientation](#).

Remarks

This method is valid only if [IMassPropertyOverrideOptions::OverrideMomentsOfInertia](#) is set to true.

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassProperty2::PrincipalAxesOfInertia Property \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetOverridePrincipalMomentsOfInertia Method (IMassPropertyOverrideOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMassPropertyOverrideOptions Interface](#) : SetOverridePrincipalMomentsOfInertia Method (IMassPropertyOverrideOptions)

Overrides the calculated principal moments of inertia of the model currently being edited.

.NET Syntax

Visual Basic (Declaration)

```
Function SetOverridePrincipalMomentsOfInertia( _
    ByVal Value As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMassPropertyOverrideOptions
Dim Value As System.Object
Dim value As System.Boolean

value = instance.SetOverridePrincipalMomentsOfInertia(Value)
```

C#

```
System.bool SetOverridePrincipalMomentsOfInertia(
    System.object Value
)
```

C++/CLI

```
System.bool SetOverridePrincipalMomentsOfInertia(
    & System.Object^ Value
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Value

Array of three doubles of the principal moments of inertia: [Px, Py, Pz]

Return Value

True if the principal moments of inertia are successfully overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [MassPropertyOverrideOptions::SetOverridePrincipalMomentsOfInertia](#).

Remarks

This method is valid only if [IMassPropertyOverrideOptions::OverrideMomentsOfInertia](#) is set to true.

See Also

[IMassPropertyOverrideOptions Interface](#)
[IMassPropertyOverrideOptions Members](#)
[IMassProperty2::PrincipalMomentsOfInertia Property\(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

IMate (Type)

IMate Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMate Interface

Obsolete. Superseded by [IMate2](#).

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IMate
```

Visual Basic (Usage)



```
Dim instance As IMate
```

C#

```
public interface IMate
```

C++/CLI

```
public interface class IMate
```

Visual Basic for Applications (VBA) Syntax

See [Mate](#).

See Also

[IMate Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IMate Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMate Interface

For a list of all members of this type, see [IMate members](#).

Public Methods

Name	Description
GetEntity	Obsolete. Superseded by IMate2::MateEntity .
GetMateDimensionValue	Obsolete. Superseded by IMate2::DisplayDimension2 .
GetMateEntities	Obsolete. Not superseded.
GetMateParams	Obsolete. Not superseded.
IGetEntity	Obsolete. Superseded by IMate2::MateEntity .
IGetMateDimensionValue	Obsolete. Superseded by IMate2::DisplayDimension2 .
IGetMateEntities	Obsolete. Not superseded.
IGetMateParams	Obsolete. Not superseded.

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See Also

IMate Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMate Interface

The following tables list the members exposed by [IMate](#).

Public Methods

Name	Description
GetEntity	Obsolete. Superseded by IMate2::MateEntity .
GetMateDimensionValue	Obsolete. Superseded by IMate2::DisplayDimension2 .
GetMateEntities	Obsolete. Not superseded.
GetMateParams	Obsolete. Not superseded.
IGetEntity	Obsolete. Superseded by IMate2::MateEntity .
IGetMateDimensionValue	Obsolete. Superseded by IMate2::DisplayDimension2 .
IGetMateEntities	Obsolete. Not superseded.
IGetMateParams	Obsolete. Not superseded.

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See Also

[IMate Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

GetEntity Method (IMate)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate Interface](#) : GetEntity Method (IMate)

Obsolete. Superseded by [IMate2::MateEntity](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetEntity( _  
    ByVal WhichOne As System.Integer _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IMate  
Dim WhichOne As System.Integer  
Dim value As System.Object  
  
value = instance.GetEntity(WhichOne)
```

C#

```
System.Object GetEntity(  
    System.int WhichOne  
)
```

C++/CLI

```
System.Object^ GetEntity(  
    & System.int WhichOne  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichOne

Visual Basic for Applications (VBA) Syntax

See [Mate::GetEntity](#).

See Also

[IMate Interface](#)
[IMate Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMateDimensionValue Method (IMate)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate Interface](#) : GetMateDimensionValue Method (IMate)

Obsolete. Superseded by [IMate2::DisplayDimension2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetMateDimensionValue() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMate  
Dim value As System.Object  
  
value = instance.GetMateDimensionValue()
```

C#

```
System.object GetMateDimensionValue()
```

C++/CLI

```
System.Object^ GetMateDimensionValue();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Mate::GetMateDimensionValue](#).

See Also

[IMate Interface](#)
[IMate Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMateEntities Method (IMate)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate Interface](#) : GetMateEntities Method (IMate)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMateEntities() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMate  
Dim value As System.Object  
  
value = instance.GetMateEntities()
```

C#

```
System.Object GetMateEntities()
```

C++/CLI

```
System.Object^ GetMateEntities();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Mate::GetMateEntities](#).

See Also

[IMate Interface](#)
[IMate Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetMateParams Method (IMate)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate Interface](#) : GetMateParams Method (IMate)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMateParams() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMate  
Dim value As System.Object  
  
value = instance.GetMateParams()
```

C#

```
System.Object GetMateParams()
```

C++/CLI

```
System.Object^ GetMateParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Mate::GetMateParams](#).

See Also

[IMate Interface](#)
[IMate Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetEntity Method (IMate)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate Interface](#) : IGetEntity Method (IMate)

Obsolete. Superseded by [IMate2::MateEntity](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEntity(  
    ByVal WhichOne As System.Integer _  
) As Entity
```

Visual Basic (Usage)



```
Dim instance As IMate  
Dim WhichOne As System.Integer  
Dim value As Entity  
  
value = instance.IGetEntity(WhichOne)
```

C#

```
Entity IGetEntity(  
    System.int WhichOne  
)
```

C++/CLI

```
Entity^ IGetEntity(  
& System.int WhichOne  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichOne

Visual Basic for Applications (VBA) Syntax

See [Mate::IGetEntity](#).

See Also

[IMate Interface](#)
[IMate Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetMateDimensionValue Method (IMate)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate Interface](#) : IGetMateDimensionValue Method (IMate)

Obsolete. Superseded by [IMate2::DisplayDimension2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMateDimensionValue() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMate  
Dim value As System.Double  
  
value = instance.IGetMateDimensionValue()
```

C#

```
System.double IGetMateDimensionValue()
```

C++/CLI

```
System.double IGetMateDimensionValue();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Mate::IGetMateDimensionValue](#).

See Also

[IMate Interface](#)
[IMate Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMateEntities Method (IMate)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate Interface](#) : IGetMateEntities Method (IMate)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetMateEntities( _  
    ByRef Entity1 As MateEntity, _  
    ByRef Entity2 As MateEntity _  
)
```

Visual Basic (Usage)



```
Dim instance As IMate  
Dim Entity1 As MateEntity  
Dim Entity2 As MateEntity  
  
instance.IGetMateEntities(Entity1, Entity2)
```

C#

```
void IGetMateEntities(  
    out MateEntity Entity1,  
    out MateEntity Entity2  
)
```

C++/CLI

```
void IGetMateEntities(  
& [Out] MateEntity^ Entity1,  
& [Out] MateEntity^ Entity2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entity1

Entity2

Visual Basic for Applications (VBA) Syntax

See [Mate::IGetMateEntities](#).

See Also

[IMate Interface](#)
[IMate Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetMateParams Method (IMate)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate Interface](#) : IGetMateParams Method (IMate)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetMateParams( _  
    ByRef MateType As System.Integer, _  
    ByRef AlignFlag As System.Integer, _  
    ByRef CanBeFlipped As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IMate  
Dim MateType As System.Integer  
Dim AlignFlag As System.Integer  
Dim CanBeFlipped As System.Integer  
  
instance.IGetMateParams(MateType, AlignFlag, CanBeFlipped)
```

C#

```
void IGetMateParams(  
    out System.int MateType,  
    out System.int AlignFlag,  
    out System.int CanBeFlipped  
)
```

C++/CLI

```
void IGetMateParams(  
& [Out] System.int MateType,  
& [Out] System.int AlignFlag,  
& [Out] System.int CanBeFlipped  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MateType

AlignFlag

CanBeFlipped

Visual Basic for Applications (VBA) Syntax

See [Mate::IGetMateParams](#).

See Also

[IMate Interface](#)
[IMate Members](#)